MODULE 1

Foundations of Business Research & Statistics

DBS-PC-BR-601-T · Business Research and Statistics · MBA Semester I

Excel-based · Anderson, Sweeney et al. Ch. 1–2 | Levine, Szabat & Stephan Ch. 1

What you will be able to do

Learning outcomes for this module

01

Explain why managers need statistics

Describe how statistical thinking supports decisions in accounting, finance, marketing, operations and HR.

02

Run a business research study end to end

Move from problem definition through design, data collection, analysis and reporting.

03

Classify data correctly

Distinguish categorical from quantitative data and assign the right measurement scale.

04

Select an appropriate sampling method

Choose between simple random, systematic, stratified and cluster sampling — and justify it.

05

Apply ethical and privacy standards

Recognise bias, misleading presentation and data-protection obligations.

06

Prepare a dataset in Excel

Build a clean, analysis-ready worksheet using validation, sorting, filtering and de-duplication.

Module roadmap

Five topics, then two case problems

1

Business research & statistics

Why it matters, the research process, key vocabulary

2

Data types & measurement scales

Categorical vs quantitative; nominal, ordinal, interval, ratio

3

Sampling techniques

Random, systematic, stratified, cluster; bias and error

4

Ethics in business research

GDPR, data privacy, bias, misleading statistics

5

Preparing data in Excel

Worksheet design, validation, cleaning, missing values

Case problems: Pelican Stores (Ch. 2) and Motion Picture Industry (Ch. 2) — both worked in Excel using the datasets from the Anderson & Sweeney companion site.

1

Business Research & Statistics

Why it matters, the research process, and the vocabulary you need

What do we mean by ‘statistics’?

The word carries two distinct meanings in business

1 Statistics as numbers

Numerical facts — averages, medians, percentages, index numbers and ratios — that summarise a business or economic situation.

“Q3 revenue rose 12%.” “The median salary is ₹8.4 lakh.”

2 Statistics as a discipline

The art and science of collecting, analysing, presenting and interpreting data so that a manager can act on it.

This is what the course teaches — with Excel as the instrument.

The manager’s question is never ‘what is the mean?’ It is ‘should we open the store, change the price, or keep the supplier?’ Statistics converts data into an answer you can defend.

Why decision makers study statistics

Four things statistics lets a manager do that intuition cannot

A

Describe

Present and describe business data and information properly, so an audience sees what is actually there.

B

Infer

Draw conclusions about large populations using information collected from a much smaller sample.

C

Forecast

Make reliable, quantified predictions about a business activity instead of guessing.

D

Improve

Monitor and improve business processes by measuring variation and reacting to it.

Source: Levine, Szabat & Stephan, Statistics for Managers Using Microsoft Excel, Ch. 1

Statistics across the business functions

Every specialisation in your MBA runs on this material

Accounting

Public accounting firms use statistical sampling when conducting audits for clients.

Finance

Advisors use price–earnings ratios and dividend yields to guide investment advice.

Economics

Economists build statistical forecasts of output, inflation and employment.

Marketing

Point-of-sale scanner data feeds pricing, promotion and assortment research.

Operations

Statistical quality control charts monitor the output of a production process.

Info. Systems

Statistical summaries help administrators assess network and system performance.

Two branches of statistics

Descriptive summarises what you have; inferential reaches beyond it

DESCRIPTIVE STATISTICS

Collecting, summarising and describing the data you actually hold.

- Tables — frequency and percent frequency distributions
- Graphs — bar charts, histograms, scatter diagrams
- Numbers — mean, median, standard deviation
- Covered in Module 2 of this course

INFERENTIAL STATISTICS

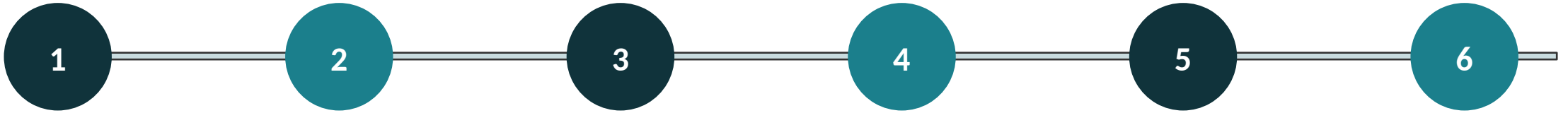
Drawing conclusions about a population from sample results alone.

- Estimation — confidence intervals for a mean or proportion
- Hypothesis testing — is the observed difference real?
- Always carries a stated level of uncertainty
- Covered in Module 3 of this course

Hudson Auto Repair (Anderson Ch. 1): 50 tune-up invoices give a sample mean parts cost of $79 — descriptive. Using that $79 to estimate the cost for all tune-ups — inferential.

The business research process

Six steps — each one constrains the next



1	2	3	4	5	6
Define the problem	**Design the study**	**Sample & collect**	**Prepare the data**	**Analyse**	**Report & act**
State the managerial decision, then the research question	Experimental or observational? What variables?	Frame, method, sample size, instrument	Clean, code, validate — in Excel	Describe, then test and model	Communicate so a decision follows

Most research failures happen at step 1 or step 3. A perfectly executed analysis of the wrong question, or of a biased sample, is worthless — no amount of Excel can repair it.

Types of research studies

How you obtain the data determines what you may claim

EXPERIMENTAL

The variable of interest is identified, then one or more other variables are controlled so you can observe their influence.

A/B test of two web-page layouts; the 1954 Salk polio vaccine trial — nearly two million children.

OBSERVATIONAL

No attempt is made to control or influence the variables of interest — you record what happens.

Customer surveys; studies of smokers vs non-smokers, where the researcher does not decide who smokes.

Exploratory

Clarify a vague problem. Interviews, focus groups, literature scan.

"What is hurting our renewals?"

Descriptive

Measure and profile. Surveys, panels, transaction data.

"Who are our promotional customers?"

Causal

Establish cause and effect. Controlled experiments.

"Did the coupon cause the extra spend?"

The vocabulary of a data set

Element, variable, observation — fixed meanings you will use all semester

Element

The entity on which data are collected — here, a company.

Variable

A characteristic of interest for the elements — here, Annual Sales.

Observation

The set of measurements obtained for one element — one row.

Data set

All the data collected in the study. n elements × k variables = n×k values.

Company	Stock Exch.	Annual Sales ($M)	Earn/Share ($)
Dataram	NQ	73.10	0.86
EnergySouth	N	74.00	1.67
Keystone	N	365.70	0.86
LandCare	NQ	111.40	0.33
Psychemedics	N	17.60	0.13

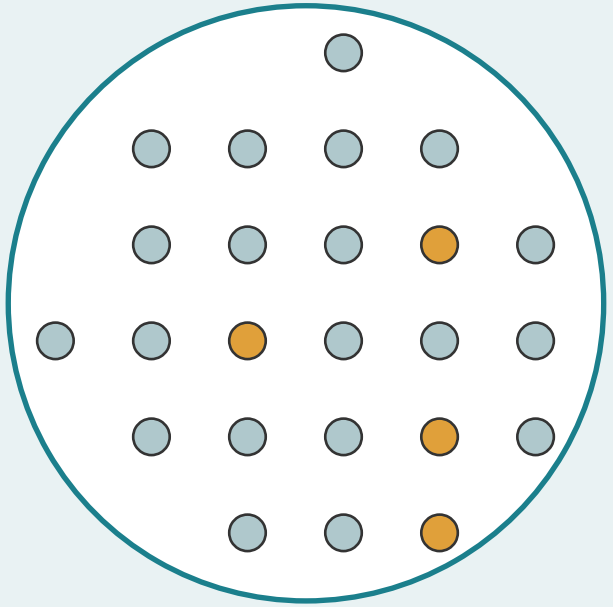
5 elements × 4 variables = 20 data values

Source: Anderson, Sweeney et al., Statistics for Business and Economics, Ch. 1

Population, sample and the inference you buy with them

Everything in Modules 3 and 4 rests on this diagram

POPULATION



All the items you want a conclusion about. A measure describing it is a PARAMETER.

sample

infer

SAMPLE



The portion of the population selected for analysis. A measure computed from it is a STATISTIC.

Census = collect data for the entire population. **Sample survey** = collect data for a sample. **Statistical inference** = using the sample to estimate and test claims about the population.

Where business data come from

Primary and secondary sources — and what each costs you

PRIMARY SOURCES

You are the data collector.

- A survey you designed and ran
- Data from an experiment you conducted
- Observation of a process or a store

SECONDARY SOURCES

Someone else collected it.

- Internal records — sales, inventory, HR, credit
- Government — MoSPI, RBI, Census, BLS, Eurostat
- Industry bodies, databases (CMIE, Bloomberg), the web

THREE THINGS TO CHECK BEFORE YOU USE ANY DATASET

Time

Searching takes time. Information may be stale by the time you have it.

Cost

Organisations charge for data even when selling data is not their business.

Errors

Data acquired carelessly, or reused because it was handy, produces misleading conclusions.

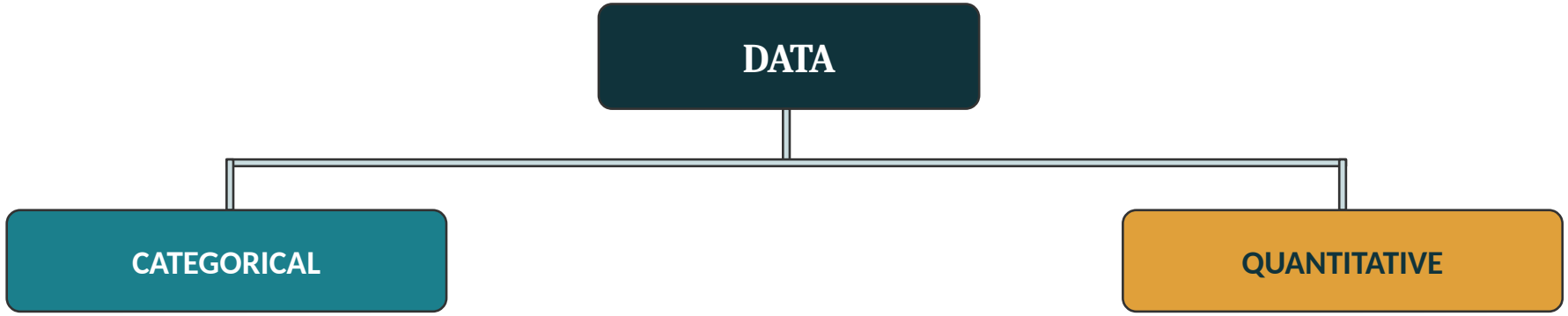
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Data Types & Measurement Scales

What kind of data you have decides what analysis is legitimate

Categorical or quantitative?

The first question to ask about any variable



Labels or names identifying an attribute. Also called qualitative. May be numeric codes or text. Uses the nominal or ordinal scale. Statistical options are limited.

- Nominal - Gender, region, brand
- Ordinal - Satisfaction rating, bond grade

Indicates how many or how much. Always numeric. Ordinary arithmetic is meaningful — so far more analysis is available.

- Discrete — counted (items purchased, defects per hour)
- Continuous — measured (net sales, weight, time on site)

The statistical analysis that is appropriate depends entirely on this split.

The four scales of measurement

Each scale adds information — and unlocks more analysis

Each level keeps everything the level below it has, and adds one property.

		EXAMPLES
RATIO	Level 4 Interval + a true zero, so ratios are meaningful.	Net sales, age, credit hours, distance, time
INTERVAL	Level 3 Ordinal + equal intervals. No true zero.	Temperature in °C; a standardised test score
ORDINAL	Level 2 Nominal + meaningful rank. Gaps not equal.	Bond rating: AAA, AA, A, BBB …
NOMINAL	Level 1 Labels only. No order.	Payment method: Discover, MasterCard, Proprietary Card

The scale determines both the summarisation and the statistical analyses that are appropriate.

Nominal and ordinal in practice

Categorical variables you will meet in the case data

NOMINAL

Classifies into distinct categories with no ranking implied. A non-numeric label or a numeric code may be used — the code carries no arithmetic meaning.

Variable	Categories
Method of payment	Discover / MasterCard / Proprietary Card
Gender	Female / Male
Marital status	Married / Single
School enrolled in	Business / Humanities / Education

Coding Business = 1, Humanities = 2 does NOT make Humanities 'twice' Business.

ORDINAL

Has all the properties of nominal data, and the order or rank of the data is meaningful. The distance between ranks is not.

Variable	Ordered categories
Class standing	Freshman → Sophomore → Junior → Senior
Product satisfaction	Unsatisfied → Neutral → Satisfied
S&P bond rating	AAA → AA → A → BBB → BB …
Faculty rank	Instructor → Assistant → Associate → Professor

The gap from Neutral to Satisfied is not necessarily the same as Unsatisfied to Neutral — so an average rating is only ever approximate.

Interval and ratio in practice

Quantitative variables — and the one property that separates them

INTERVAL

Ordinal properties plus a fixed unit of measure, so differences are meaningful. There is no true zero, so ratios are not.

Melissa scores 1985 on the SAT, Kevin 1880.

Valid: Melissa scored 105 points more than Kevin.
Invalid: Melissa is 1.06 times as able as Kevin.

RATIO

All the properties of interval data plus a true zero point meaning 'none of the variable exists' — so ratios are meaningful.

Melissa has 36 credit hours, Kevin has 72.

Valid: Kevin has twice as many credit hours as Melissa.
Distance, height, weight, time, sales and cost are all ratio.

WHAT EACH SCALE PERMITS

Scale	Count / mode	Rank / median	Add & subtract / mean	Multiply & divide / ratios	Typical chart
Nominal	Yes	No	No	No	Bar, pie, Pareto
Ordinal	Yes	Yes	No	No	Bar, stacked bar
Interval	Yes	Yes	Yes	No	Histogram, line
Ratio	Yes	Yes	Yes	Yes	Histogram, scatter, box plot

Cross-sectional and time series data

When the data were collected changes what you can say with them

CROSS-SECTIONAL

Collected at the same, or approximately the same, point in time.

Building permits issued in November 2012 in each county of Ohio. Or: the 100 Pelican Stores transactions recorded on one day.

TIME SERIES

Collected over several time periods.

- Understand what happened in the past
- Identify trends and seasonal patterns
- Project future levels — Module 4

A retail time series: trend plus a Q4 seasonal peak



Illustrative data

3

Sampling Techniques

How you pick the sample decides whether the answer means anything

Why sample instead of taking a census?

And the four terms you must be able to define

Cost

A census of every customer is usually unaffordable; a sample of 400 is not.

Time

By the time a census is complete, the decision has already been made.

Feasibility

Testing is sometimes destructive — you cannot crash-test every car you sell.

Accuracy

A small, carefully collected sample often beats a large, sloppy census.

Target population

The complete set of items you want the conclusion to apply to.

Sampling frame

The actual list you draw from. Gaps between frame and population create bias.

Sampling error

The natural difference between a sample result and the population value.

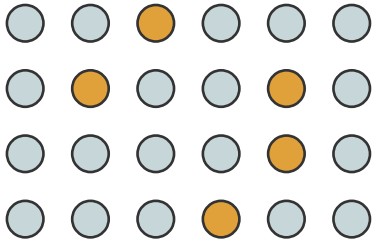
Non-sampling error

Bad questions, non-response, data-entry mistakes — a bigger sample will not fix these.

Four probability sampling methods

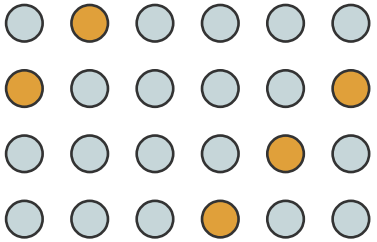
Every member of the population has a known, non-zero chance of selection

Simple random



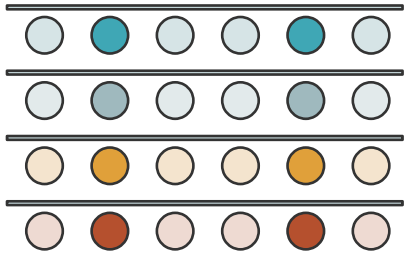
Every possible sample of size n is equally likely. Draw with Excel: RAND() then sort, or the Analysis ToolPak's Sampling tool.

Systematic



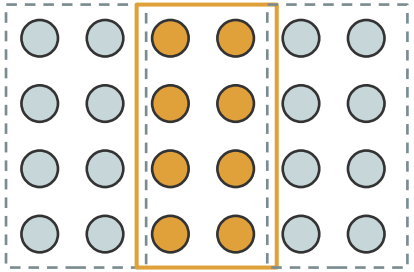
Pick a random start, then take every k-th item, where k = N/n. Fast for long lists and production lines.

Stratified



Split the population into homogeneous strata, then sample randomly within each. Gains precision.

Cluster



Split into naturally occurring clusters, randomly choose whole clusters, survey everyone inside. Cheapest.

Highlighted dots = selected units. *Source: Anderson, Sweeney et al. Ch. 7 'Other Sampling Methods'.*

Choosing a sampling method

Match the method to the population structure and the budget

Method	How it works	Use it when	Watch out for
Simple random	Number the frame 1…N; draw n numbers at random	The population is fairly homogeneous and a full list exists	Needs a complete frame; can miss small subgroups
Systematic	Random start, then every k-th unit, k = N/n	The list is long and ordered arbitrarily	Hidden periodicity in the list biases the sample
Stratified	Divide into strata, sample randomly inside each	Subgroups differ from each other but are similar inside	You must know stratum sizes in advance
Cluster	Divide into clusters, select whole clusters at random	The population is geographically spread and travel is costly	Less precise per unit; clusters must be internally varied

Non-probability sampling

Convenience, judgment, quota and snowball samples. Fast and cheap, and common in student projects — but the selection probabilities are unknown, so no valid confidence interval or hypothesis test can be computed from them.

Three biases to name in your report

Selection bias — the frame excludes part of the population. Non-response bias — those who reply differ from those who do not. Measurement bias — the question itself pushes the answer.

4

Ethics in Business Research

Bias, privacy and honest presentation — the parts nobody audits until it is too late

Where statistical practice goes wrong

Unethical behaviour in a statistical study takes five recurring forms

1	**Improper sampling**	Choosing a frame or method that quietly excludes the people who would disagree.
2	**Inappropriate analysis**	Running a test whose assumptions the data plainly violate, then reporting the p-value anyway.
3	**Misleading graphs**	Truncated axes, missing baselines, 3-D effects, unequal category widths.
4	**Inappropriate summaries**	Reporting a mean when the distribution is badly skewed and the median tells a different story.
5	**Biased interpretation**	Presenting only the results that support the decision already taken.

The standard you are held to

Be fair, thorough, objective and neutral as you collect, analyse and present data — and be alert to the possibility of unethical behaviour by others whose numbers you consume.

The American Statistical Association's 'Ethical Guidelines for Statistical Practice' sets out 67 guidelines across eight areas — from professionalism to responsibilities to research subjects.

Data privacy: GDPR and India's DPDP Act

If your project touches personal data, these obligations apply to you

Lawfulness

Have a legal basis — usually informed consent — before you collect anything.

Purpose limitation

State why you are collecting the data; do not reuse it for a different purpose later.

Data minimisation

Collect only the fields your analysis actually needs. Do not ask for a phone number you will never call.

Accuracy

Keep records correct and up to date; correct them when a subject asks.

Storage limitation

Delete the raw data once the study is complete and the retention period ends.

Integrity & security

Protect the file. Anonymise or pseudonymise identifiers before you share a worksheet.

Rights the data subject holds

Access · rectification · erasure · restriction of processing · portability · objection · and rights relating to automated decision-making.

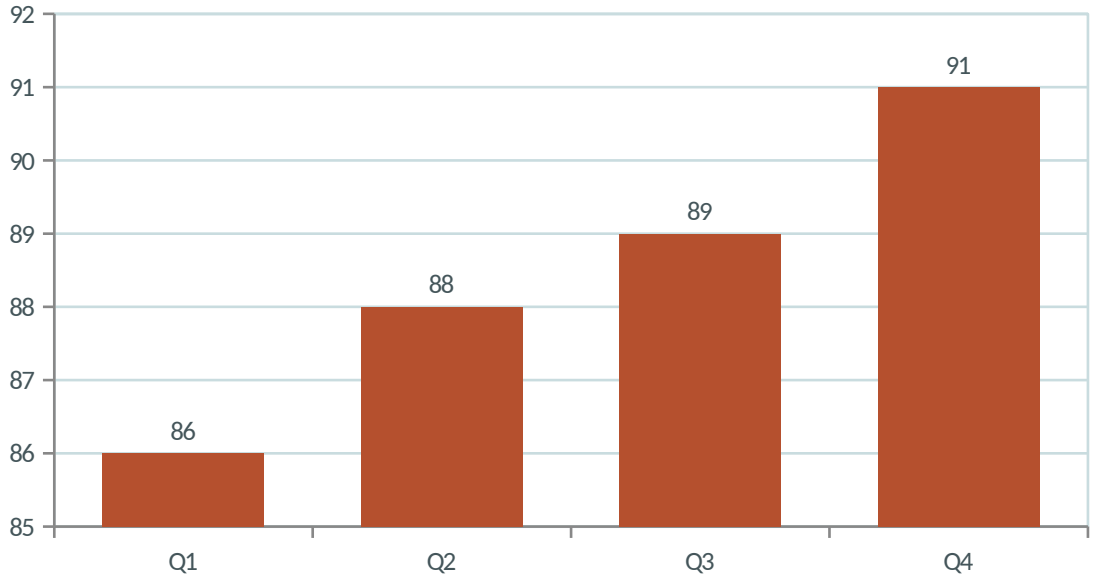
In India

The Digital Personal Data Protection Act, 2023 applies similar principles — notice, consent, purpose limitation, and the rights to access, correct and erase. Verifiable parental consent is required for anyone under 18.

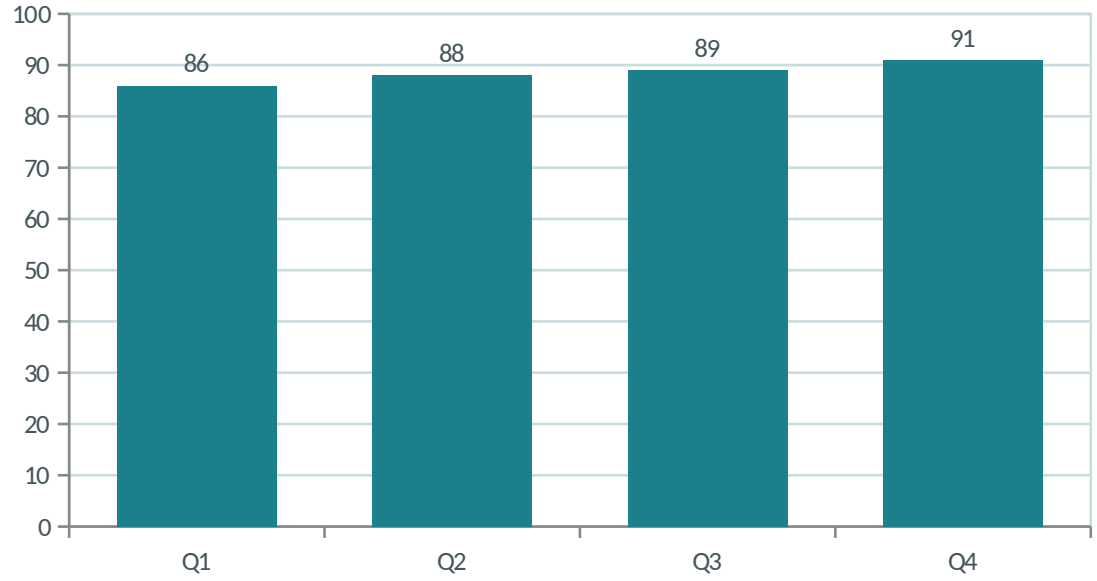
How a chart lies

Same numbers, two axes — and two completely different board meetings

Truncated axis: ‘share is surging’



Zero baseline: ‘share is flat’



Start value axes at zero for bar and column charts

Use a relative basis percentages, not raw counts, when groups differ in size

Strip the chart junk no 3-D, no gradients, no decorative clutter

Label the axes and the source every time, without exception

5

Preparing Data in Excel

A clean worksheet is not housekeeping — it is the first analytical step

The anatomy of a worksheet

Four terms, then the one layout rule that makes everything else work

Workbook

The file. A collection of worksheets and chart sheets.

Worksheet

One grid of columns and rows. Keep raw data and results on separate sheets.

Cell

The intersection of a column and a row — B4.

Cell range

A contiguous rectangular block — B4:E20. This is what every Excel tool asks you for.

Customer	Type	Items	Net Sales	Payment	Gender	Age
1	Regular	1	39.50	Discover	Male	32
2	Promotional	1	102.40	Proprietary Card	Female	36
3	Regular	1	22.50	Proprietary Card	Female	32
4	Promotional	5	100.40	Proprietary Card	Female	28
5	Regular	2	54.00	MasterCard	Female	34

One variable per column · one observation per row · no blank rows

Rows 1–5 of the PelicanStores dataset (Anderson & Sweeney, Table 2.19)

This shape — tidy, rectangular, one header row — is what PivotTables, the Analysis ToolPak and every chart tool expect. Break it and nothing downstream works.

Designing an effective worksheet

Five habits that will save you in the capstone project

1	**Associate column ranges with variables**	One column = one variable, with a single descriptive header in row 1.
2	**Never skip rows as you enter data**	Blank rows truncate ranges silently — your mean is then computed on half the data.
3	**Separate data from results**	Raw data on one sheet, calculations and output on another. Never overwrite the source.
4	**Keep the calculation chain visible**	A reader should be able to trace every number back to the starting data.
5	**Keep two versions**	One laid out for the screen, one for the printer or the report.

Red flags in a submitted workbook

- Merged cells inside the data range
- Numbers stored as text (left-aligned, green triangle)
- Units or symbols typed into the cell: ‘₹ 45,000’
- Two headers stacked in rows 1 and 2
- Colour used to encode meaning, with no column recording it
- Totals sitting in the middle of the data

The Excel data-cleaning toolkit

Where each command lives and what it fixes

1 Data Validation

Data › Data Tools

Restricts what can be typed into a cell — a list, a date range, a whole number between limits. Stops bad data at entry.

2 Remove Duplicates

Data › Data Tools

Deletes repeated records. Always take a copy of the sheet first — it is not reversible after saving.

3 Sort & Filter

Data › Sort & Filter

Orders records and isolates subsets. Filtering is how you inspect one customer type at a time.

4 Text to Columns

Data › Data Tools

Splits a single column into several, and forces text-formatted numbers back into real numbers.

5 TRIM · CLEAN · PROPER

Formulas › Text

Strips stray spaces and non-printing characters, and normalises capitalisation in category labels.

6 Find & Replace

Home › Editing

Standardises inconsistent labels — ‘M’, ‘Male’ and ‘male’ collapsed into one category.

Rule: clean on a copy, keep the original raw file untouched, and record every cleaning decision in a notes sheet.

Handling missing values

Four options — and the reasoning you must record for whichever you pick

Leave it blank

Excel’s statistical functions ignore truly empty cells. Never type 0 or ‘NA’ into a numeric column — 0 is a value and will drag your mean down.

Delete the record

Simple and defensible when very few records are affected. Dangerous if the missing values are concentrated in one group — that is how bias enters.

Impute a value

Substitute the column mean or median, or a value predicted from other columns. Reduces variability, so say so in the report.

Return to source

The best option when it is available: go back to the record or the respondent and recover the true value.

Ask why the value is missing before you decide what to do with it. A blank because the question did not apply is a different problem from a blank because the respondent refused — and only the second one threatens your conclusions.